

Hydroponic System – Final Presentation

Team 5: HydroGrow

Fall 2025

Team members:

Maung Maung – Growth Area Monitoring

Nelson Alonso – Water Flow system

Yaniss Bantse – Nutrient Water

Cameron Morgan – Growth Area Design

Each member developed a subsystem and we worked together to integrate them into a functional prototype.

Project Goal

Design a hydroponic system that maintains the proper flow of water, nutrients, and environmental conditions to allow for healthy plant growth in an indoor environment.

The client requires an educational hydroponics system that is safe, compact, economical, and allows students to observe plant growth with automated control.

Design Requirement s

Low cost

Safe for classroom use

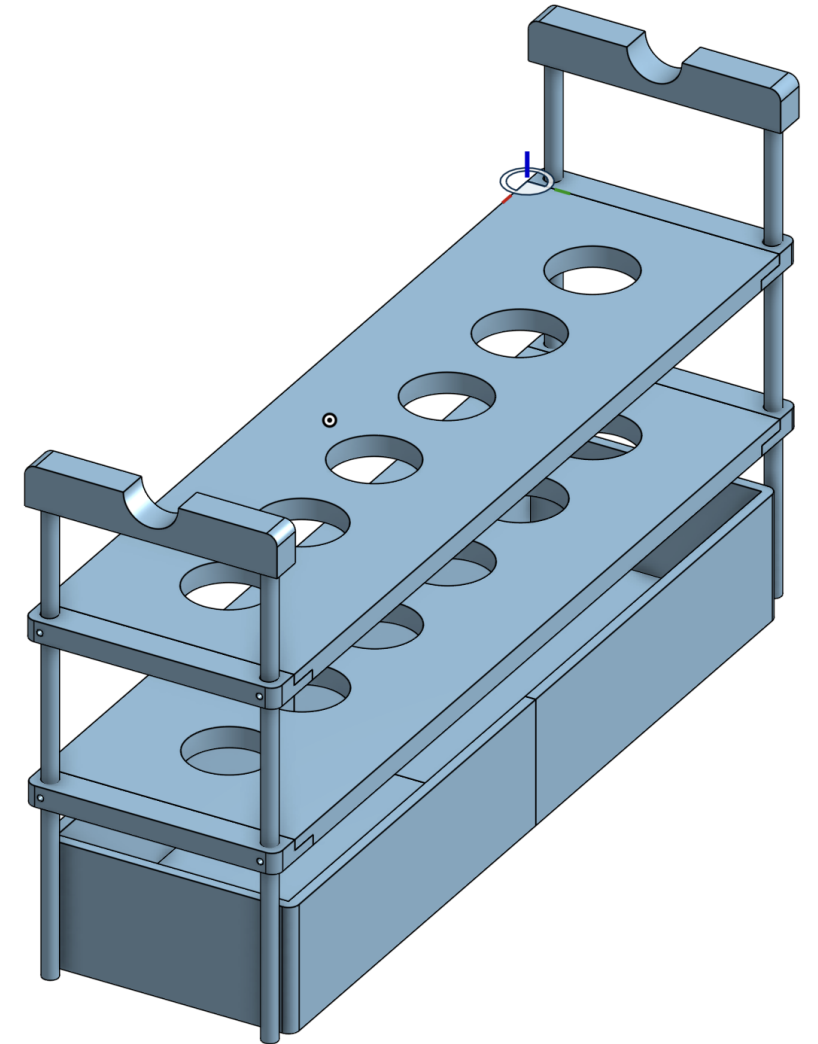
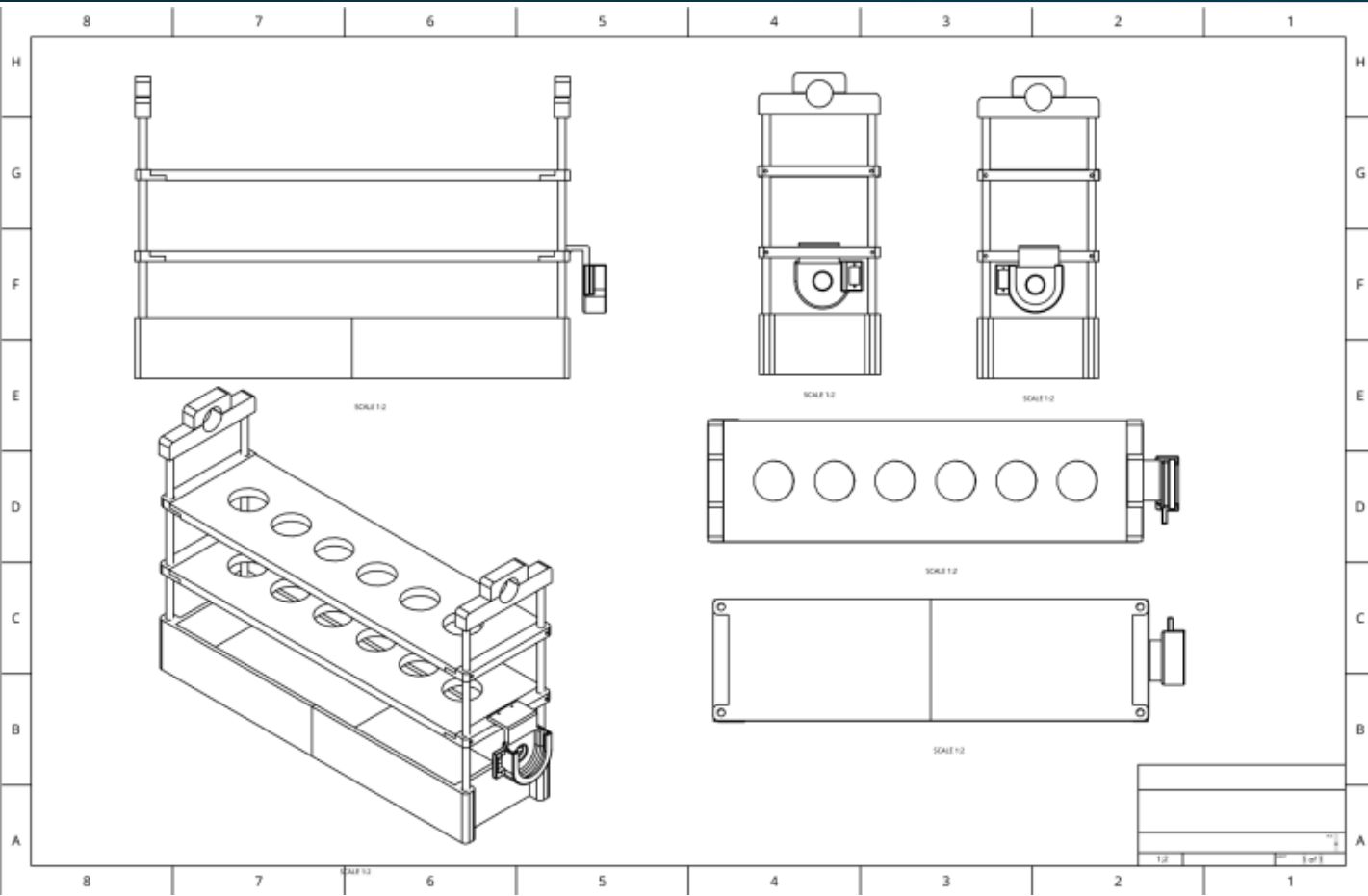
Stable water flow

Real-time monitoring

User-friendly interface

Full subsystem integration

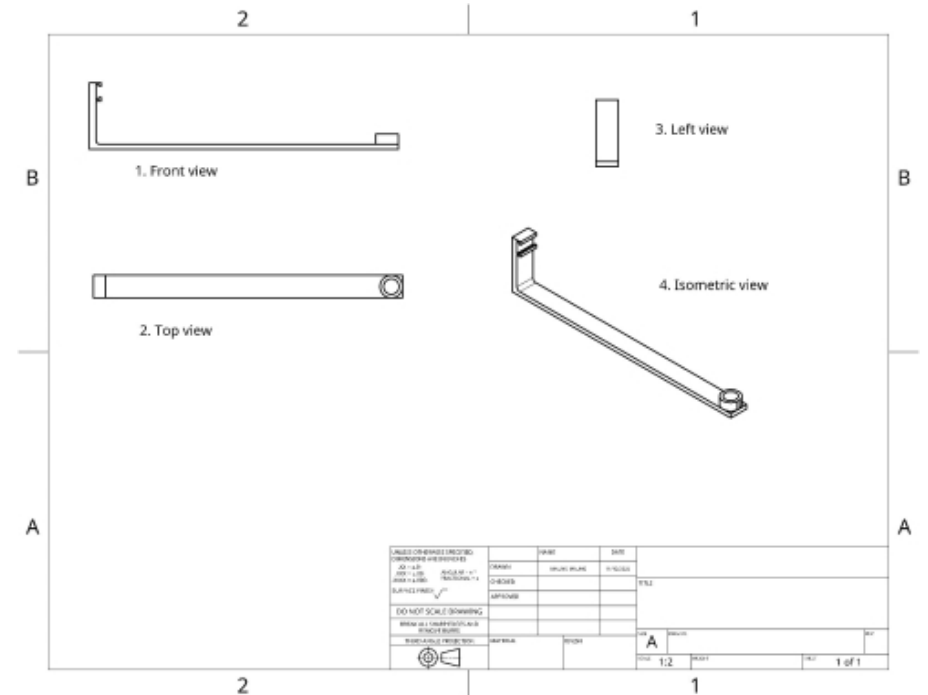
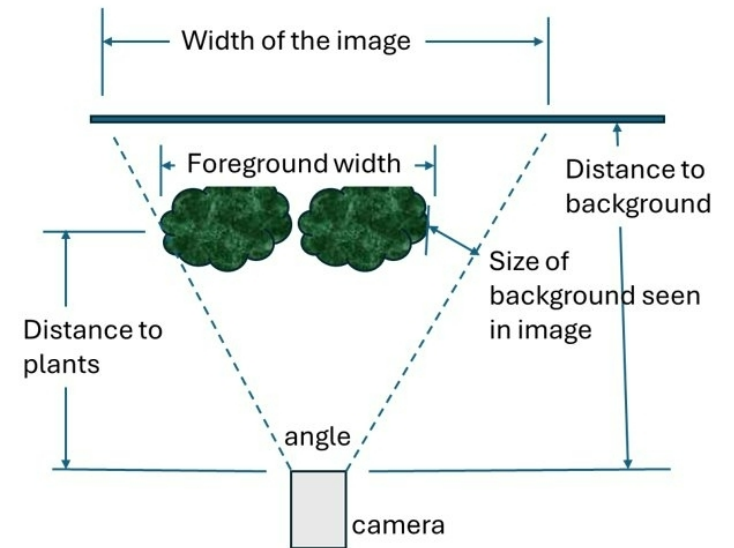
Overall System Concept



Growth Area Monitoring

The goal is to create a simple growth area monitoring system for a small hydroponic garden in a California backyard. The system should monitor plant growth stages (for example, from leaves to flowers or fruit) and measure sunlight intensity and duration using a camera and sensors. This helps the customer automatically adjust nutrient levels and lighting conditions without needing to constantly check the plants.

This 3D model holds the camera and sensors above the plants, measuring height, light levels, and growth stages automatically daily.



Water Flow System

Responsible for moving nutrient-rich water from the tank to the plant area with a controlled and stable flow rate, ensuring plants receive consistent hydration.

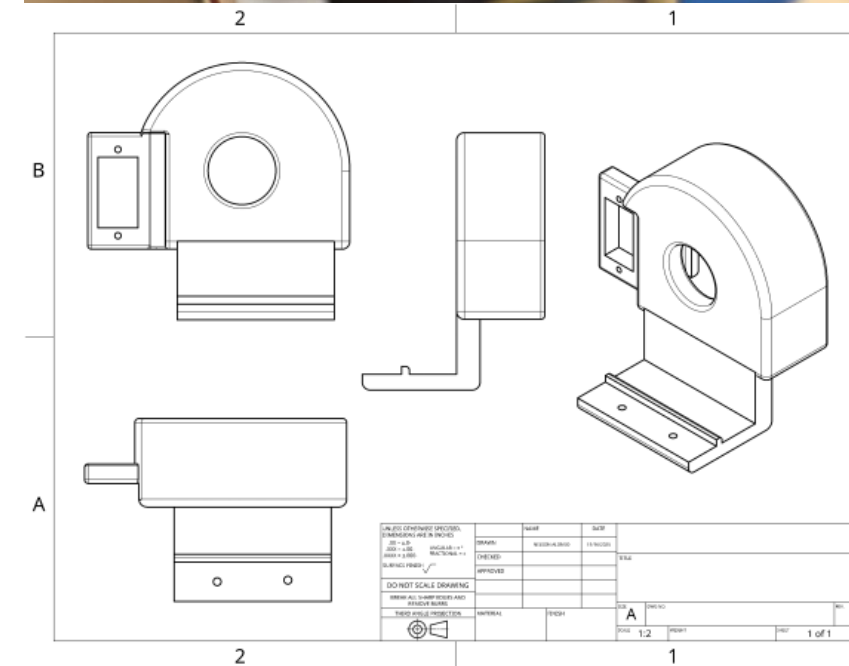
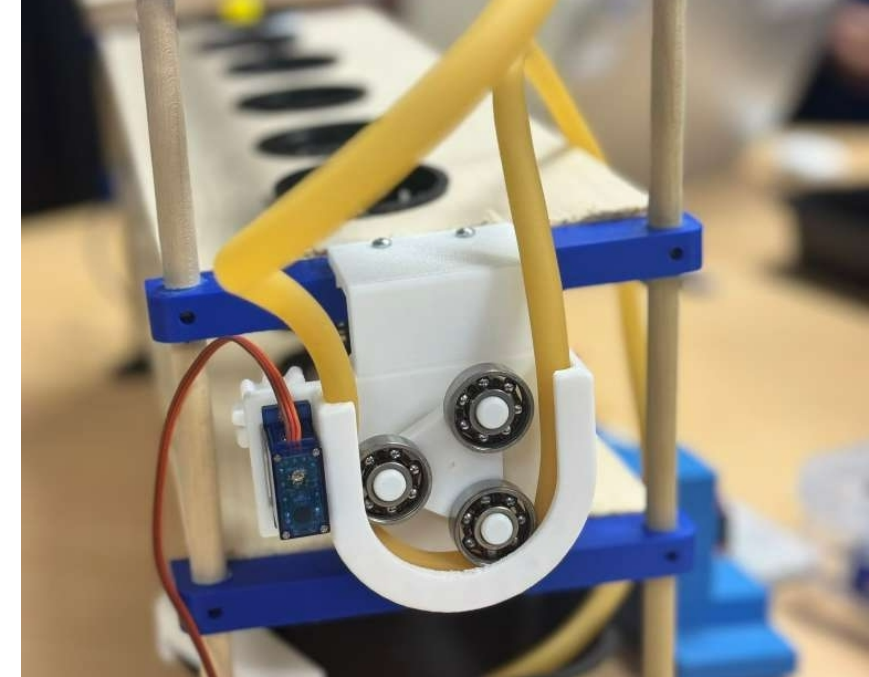
Function: Moves water through the system using a controlled flow.

Design: 3D-printed peristaltic pump + servo motor.

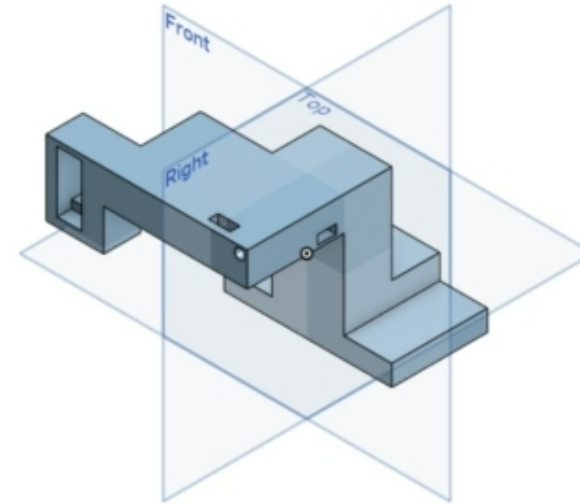
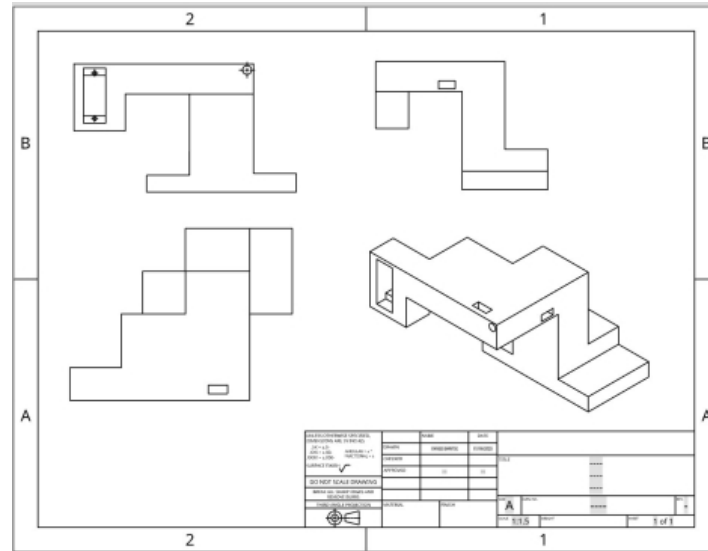
Performance: Stable flow rate, no leaks, reliable operation.

Improvements: Switched to higher-torque servo; optimized roller spacing.

Integration: Connects tank → tubing → plant area.



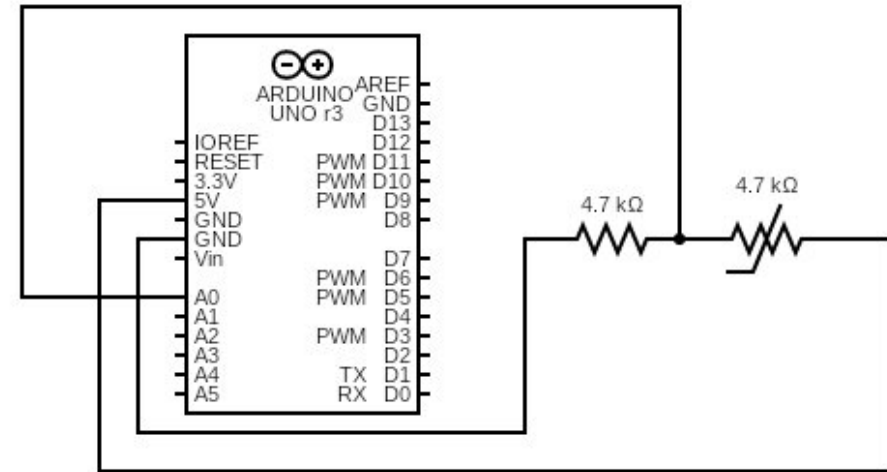
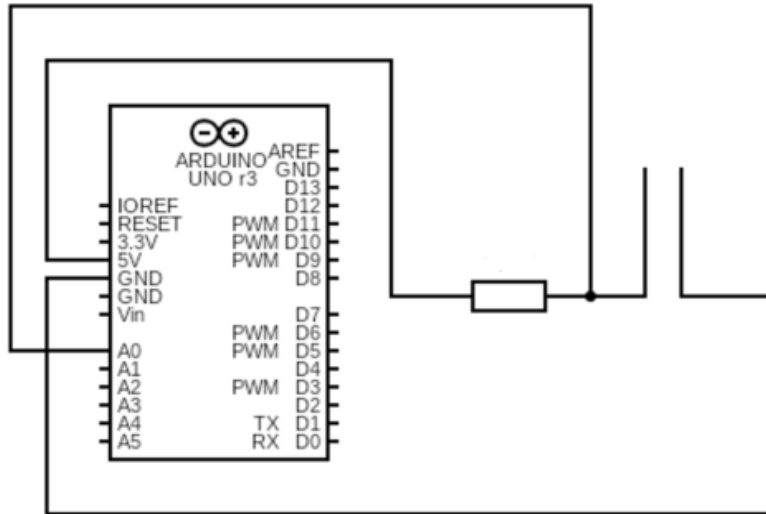
Nutrient Water



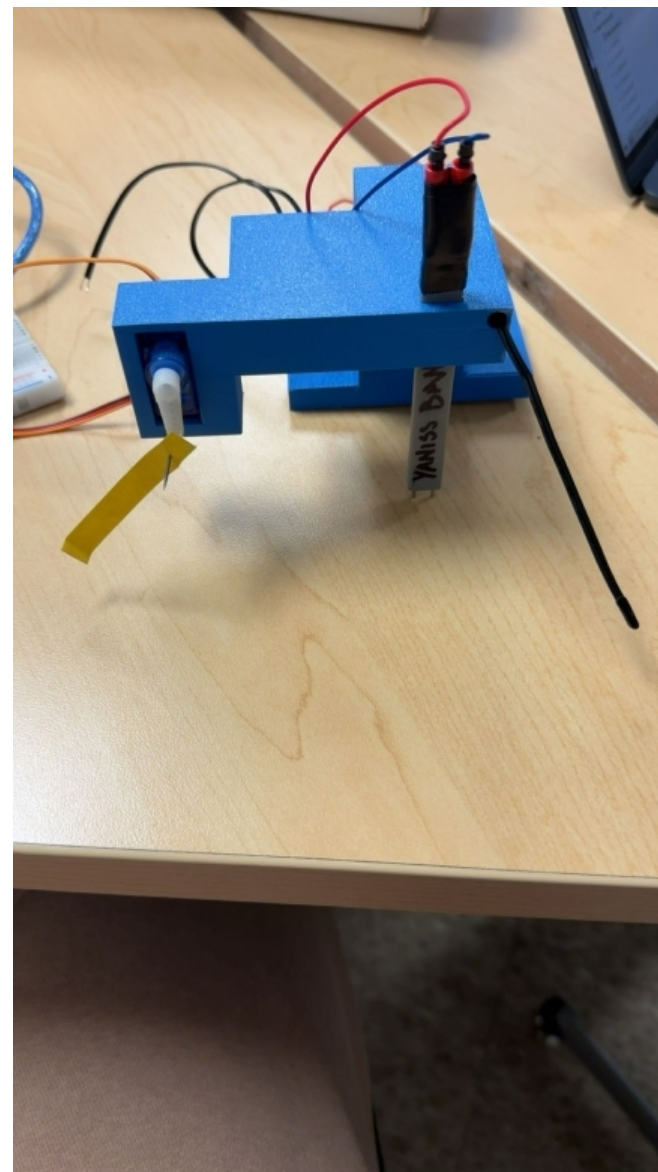
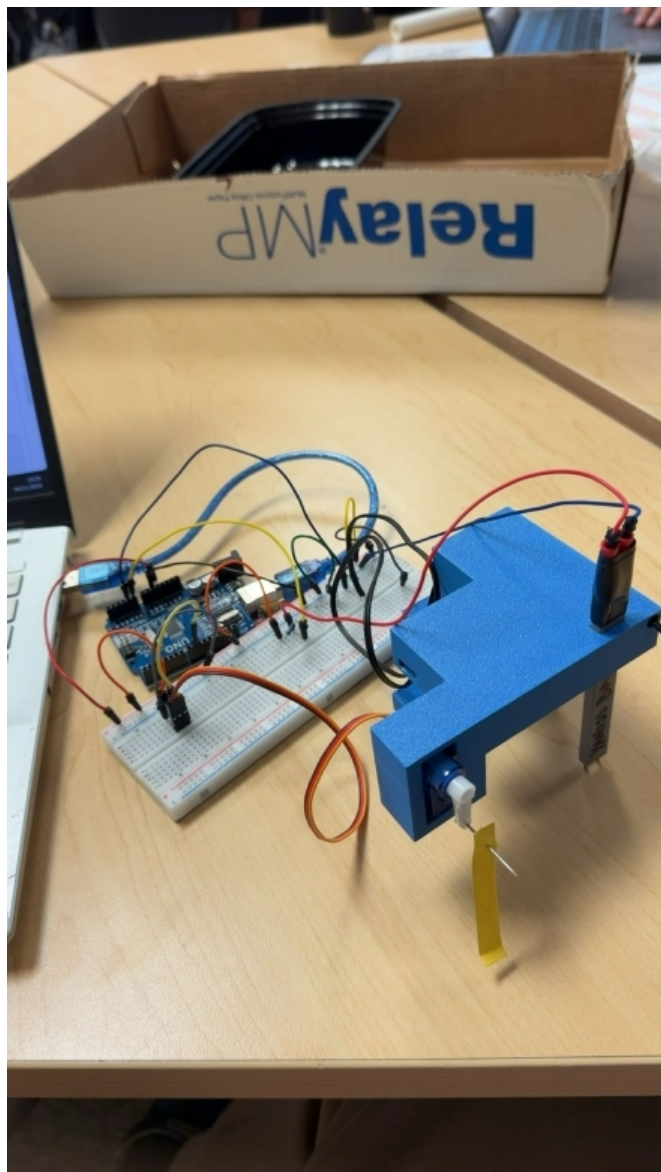
The Nutrient Water Quality subsystem monitors the chemical condition of the hydroponic water to ensure plants receive the correct amount of nutrients. Since hydroponics delivers nutrients directly through water instead of soil, maintaining proper water quality is essential for healthy growth. This subsystem mainly measures how conductive the water is, because conductivity increases as more nutrient ions are dissolved.

This 3D-printed mount holds the conductivity water sensor, the servo motor, and the waterproof thermistor. It keeps all components stable, aligned, and properly positioned to ensure accurate measurements and reliable operation of the nutrient water subsystem.

Flowchart and Schematic



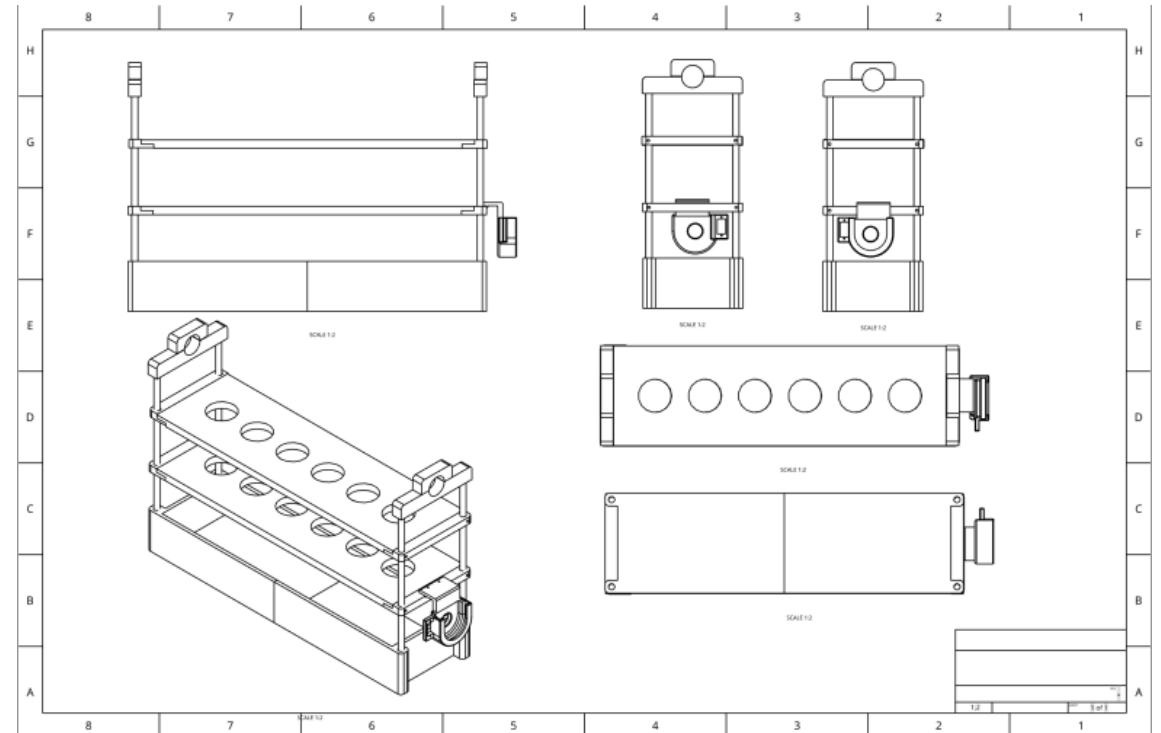
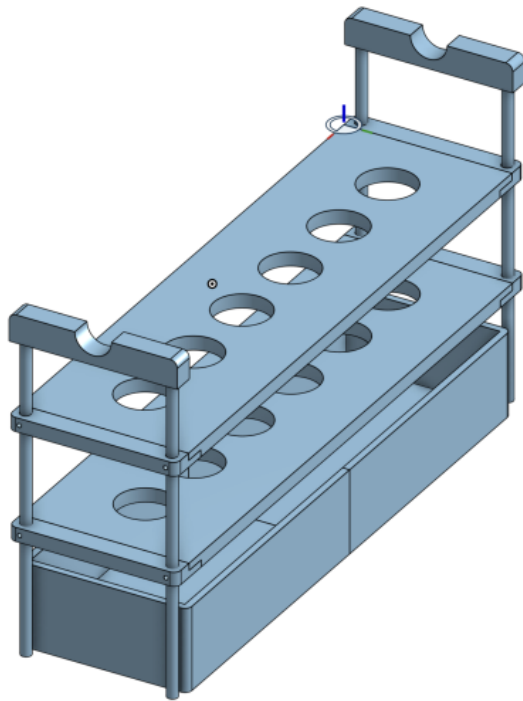
These diagrams show the schematics for the waterproof thermistor and the conductivity sensor used to measure the voltage across the water.



Growth Area Design

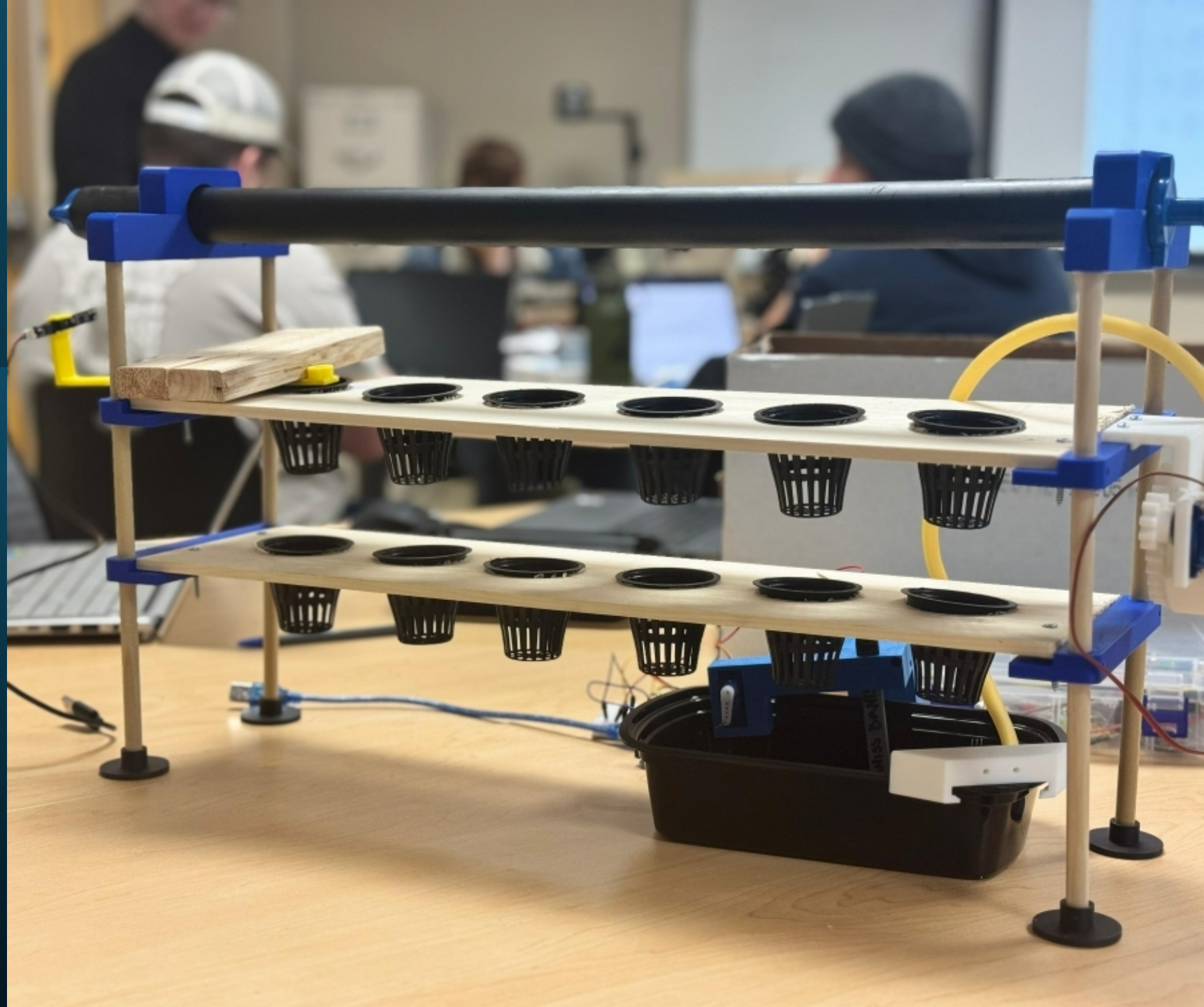
The primary goal of the growth area design is to have a functional design area for plant growth that is structurally sound and have the ability have other subsections be attached with ease.

The 3-D printed brackets on the side will allow for attachments of other subsections, the top brackets allow for a pipe to be installed for the drip system for the plant.



This is what our final
working prototype looks
like.

Final Assembled Device



Meeting Customer Needs

Low cost → 3D printed parts

Easy to use → Clear GUI

Safe → Low voltage, water contained

Portable → Compact design

Economic Analysis

Material Costs:

Servo - \$15

Tubing - \$10

PLA - \$20

Arduino - \$20

NRE – 20h x \$40 = \$800

Total Cost: \$865

Gantt Chart

Weeks 1-2: CAD

Week 3:
Fabrication

Week 4:
Assembly

Week 5: Testing

Week 6:
Integration

Week 7:
Adjustments

Future Steps



Improve sensor
integration



Refine structure
stability



Enhance GUI
features



Full system
calibration

Conclusion

System integrates pump, sensors, structure, and GUI effectively

Meets customer requirements for safety, cost, and usability

Testing shows stable flow, reliable sensor data, and successful subsystem integration

Team developed strong engineering, design, and troubleshooting skills

Next improvements: calibration, structural reinforcement, and GUI upgrades